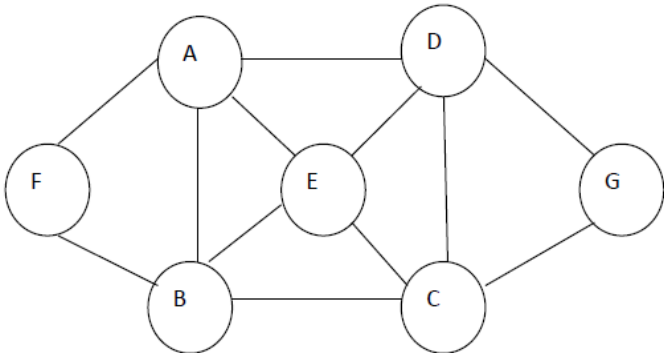
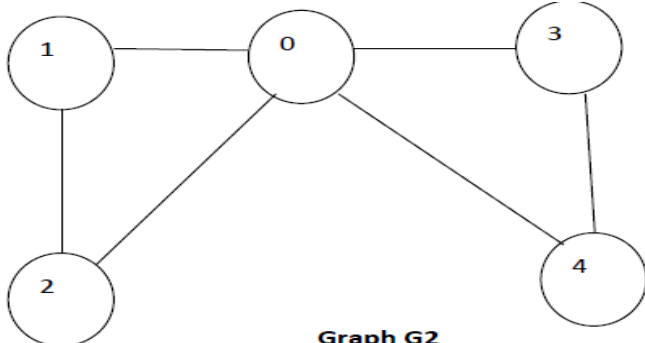
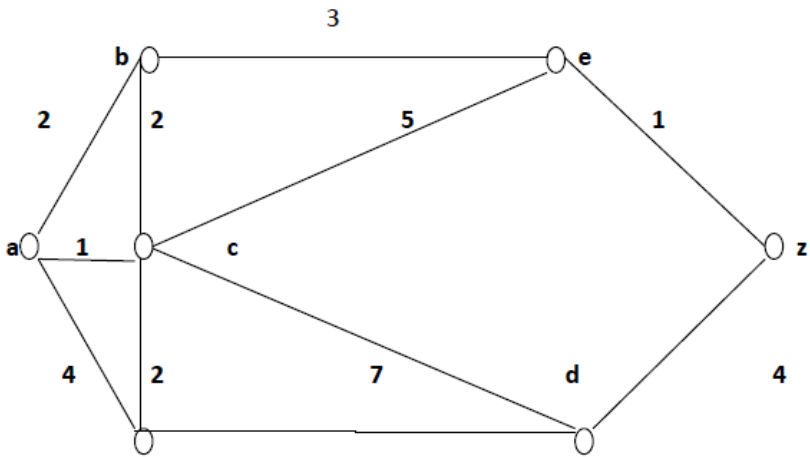


	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch: Computer Science and Engineering Semester: III Subject Code & Name: BTCOC302_Discrete Mathematics Max Marks: 60 Date: Duration: 3 Hr.			
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks	
Q. 1	Solve Any Two of the following.			
A)	Explain with example, notation used and mathematical expression to describe following terms 1)Membership 2) Subset 3) Equality of two sets	(L1/CO1)	06	
B)	Construct a truth table for each of these compound propositions and identify type of compound statements a. $(p \wedge q) \rightarrow (p \vee q)$ b. $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$	(L3/CO1)	06	
C)	Test the validity of following statement: If there is strike by student, then exam will be postponed. Exam was not postponed. Therefore, there were no strikes by students	(L3/CO1)	06	
Q.2	Solve Any Two of the following.			
A)	Let A= {1,2,3,4,6,8,12} and R be the partial order on A defined by aRb if a divides b. Determine the relational matrix for R. Construct directed graph G on A. Draw Hasse diagram of Poset (A, R).	(L1/CO2)	06	
B)	State Pigeon hole principle. If five colors are used to paint 26 doors show that at least Six doors will have the same color	(L3/CO2)	06	
C)	Let $f(x)=2x+3$, $g(x)=3x+4$, $h(x)=4x$ for $x \in \mathbb{R}$, where $\mathbb{R}$ is set of real numbers. Find gof , fog , foh and hof .	(L1/CO2)	06	
Q. 3	Solve Any One of the following.			
A)	Show that the maximum number of edges in a simple graph having n vertices is $n*(n-1)/2$.	(L1/CO3)	06	
B)	Define Euler graph and Hamiltonian graph 1) For a given graph G1 a) Find a Hamiltonian path that begins at A and ends at E. b) Find a Hamiltonian circuit that starts at A and ends with the pair of vertices E, A	(L1/CO3)	06	

	c) Find a Hamiltonian path that begins at F and ends at G. 2) For a given graph G2 find Eulerian path and Eulerian cycle  Graph G1  Graph G2		
C)	Find a shortest path between vertices a and z in the given graph using Dijkstra shortest path algorithm. 	(L3/CO3)	06
Q.4	Solve Any Two of the following.		
A)	Define the following with reference to tree with example a) Level and Height of a tree b) M-ary tree c) Fundamental Cut Sets	L1/CO3	06
B)	Construct a minimum spanning tree (MST) for the given graph using Kruskal's.	L3/CO3	06

C)	Show that a tree with n vertices has $n-1$ edges	L1/CO3	06
Q. 5	Solve Any One of the following.		
A)	Define the following terms 1. Algebraic structures 2. Semi Group 3. Monoids 4. Ring 5. Field 6. Group	L1/CO4	06
B)	For each of following, determine whether the binary operations $*$ is commutative or associative? i) N is the set of natural numbers and $a*b=a+b+2$ for all a,b from N ii) on N where $a*b=\min(a, b+2)$ iii) on R where $a*b=a^b$	L3/CO4	06
C)	Determine whether $(Z, +, *)$ is a ring with the binary operations $x + y = x + y - 7$ And $x*y = x + y - 3xy$ for all $x, y \in Z$	L1/CO4	06
	*** End ***		

The grid and the borders of the table will be hidden before final printing.